Assignment-4.4

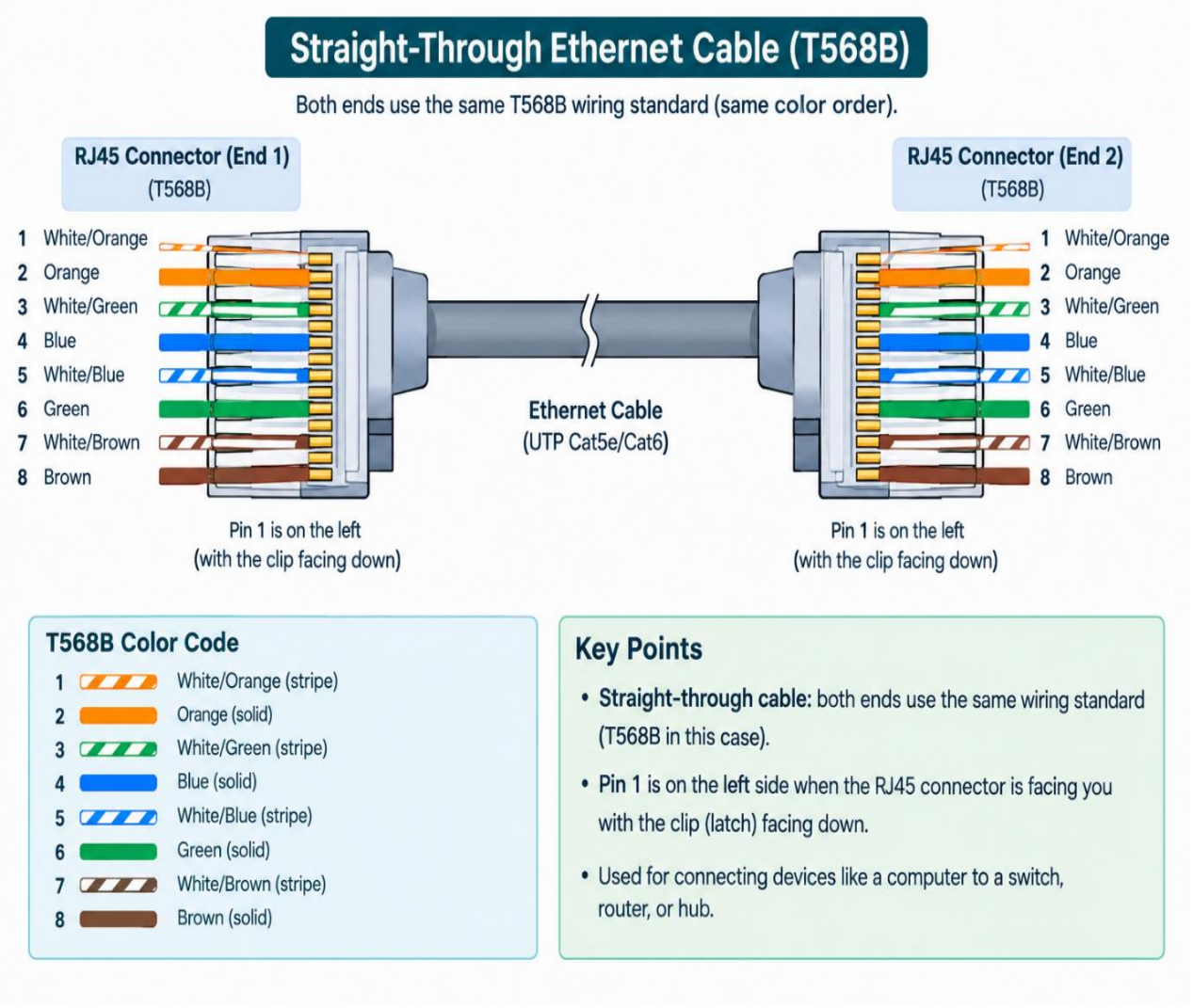
TASK-1

List 3 key differences between Ethernet cables and fiber optic cables, and give one real-world example where each type would be used (for example, home WiFi router, data center, college campus backbone).

Difference	**Ethernet Cable**	**Fiber Optic Cable**
Data Transmission	Uses electrical signals to transmit data.	Uses light signals to transmit data.
Speed and Distance	Commonly used for short distances, such as within a home or office.	Supports very high speeds over long distances.
Cost and Installation	Usually cheaper and easier to install.	Usually more expensive and requires specialized equipment for installation.
Real-World Examples	Connecting a computer to a home Wi-Fi router using a LAN cable.	Connecting network switches across different buildings on a college campus to provide high-speed connectivity.

TASK-2

Draw and label a diagram showing the internal wiring of a straight-through Ethernet cable (use T568B wiring standard), indicating the color of each wire and its pin number.



TASK-3

Watch a YouTube video demonstrating how to use an RJ45 crimper, then write a step-by-step guide (in your own words) on how to attach an RJ45 connector to a CAT6 cable.

1. Gather Your Tools and Materials

- CAT6 Cable of your desired length
- RJ45 Connectors (standard or pass-through style)
- Strain Relief Boots (optional, slide onto the cable first if using)
- RJ45 Crimping Tool (with built-in stripper and cutter)
- Network Cable Tester

2. Strip the Outer Jacket

- Slide your strain relief boot onto the cable now if you are using one.
- Use the stripping blade on your crimper or a wire stripper to remove about 1 to 1.5 inches (2.5 to 4 cm) of the outer PVC jacket from the CAT6 Cable.
- Be careful not to nick or cut the inner copper wires.
- If your cable has a central plastic spline or separator (common in CAT6) and a drain wire, cut away the plastic spline and snip or fold back the drain wire.

3. Untwist and Arrange the Wires

- Separate the four twisted pairs into eight individual wires.
- Untwist the wires as close to the cable jacket as possible—keeping twists tight helps maintain high-speed network performance.
- Arrange the eight wires side-by-side following the standard T568B color scheme (the most common layout for home and office networks):
    1. White/Orange (Orange-White)
    2. Orange
    3. White/Green (Green-White)
    4. Blue

5. White/Blue (Blue-White)
6. Green
7. White/Brown (Brown-White)
8. Brown

4. Trim the Wires Evenly

- Hold the aligned wires tightly between your thumb and forefinger.
- Use the cutter on your crimper to snip the wire ends straight across.
- Leave about 1/2 inch (1.2 to 1.5 cm) of exposed wire sticking out past the edge of the outer jacket. This length ensures the inner wires reach the gold pins while the outer jacket sits securely inside the back of the plug.

5. Insert Wires into the RJ45 Connector

- Hold the RJ45 Connector with the small plastic clip facing downward (or the gold contact pins facing up toward you).
- Gently push the flat, aligned wires straight into the connector until all eight copper tips hit the front wall.
- Look at the clear plastic end to verify that every wire is pushed all the way to the top.
- Check that the outer cable jacket extends slightly inside the base of the connector so the crimp tab clamps onto the jacket rather than the fragile individual wires.

6. Crimp the Connector

- Place the prepared plug into the RJ45 Crimping Tool socket.
- Squeeze the handles firmly and completely until the ratcheting mechanism releases. This action pushes the gold contact teeth down into the wires and locks the strain tab onto the outer jacket.
- If you are using pass-through connectors, the tool will automatically slice off the excess wire tails sticking out of the front.

7. Test the Cable

- Repeat the entire process for the other end of the cable (using T568B on both ends for a standard patch cable).
- Plug both ends into your Network Cable Tester.
- Turn the tester on and verify that all 8 indicator lights light up sequentially and match on both sides without any short circuits or missing wires

TASK-4

Imagine you are setting up a gaming café with 10 PCs for LAN gaming. Decide which cable type (Ethernet or fiber optics) and which tool (RJ45 crimper or fiber cleaver) you would use, and explain your choices in 3-4 sentences.

For a 10-PC gaming café, Ethernet cable (Cat6 or Cat6a) and an RJ45 crimper are the best choices. Ethernet easily delivers the necessary 1 Gbps to 10 Gbps speeds over short, indoor distances at a fraction of the cost of fiber optics. Fiber optics would be overkill, as its main benefits—long-distance data transmission and immunity to outdoor interference—are unnecessary for a small, single-room LAN setup. Using an RJ45 crimper allows you to quickly, affordably, and reliably custom-terminate these Ethernet cables to keep your setup clean and organized.